Semantic Monitoring for Adaptive Compute Allocation in Edge Robotic Vision-Language Systems

Devarakonda Shanmukha Sai (MHC2024013)

Department of Information Technology
Indian Institute of Information Technology, Allahabad

Supervisor: **Prof. G. C. Nandi**

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Outline

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1.1 Background and Motivation

- **Vision-Language Models (VLMs)** give edge robots open-vocabulary scene understanding.
- But each VLM call is expensive: hundreds of ms, GBs of memory, high energy.
- Robot video is **temporally redundant** — consecutive frames are semantically near-identical.
- Mainstream systems still call the VLM every frame (or every K), re-deriving the same understanding.

The opportunity

Pixel change is frequent and cheap to measure; *semantic change* is rare and is what actually warrants reasoning.

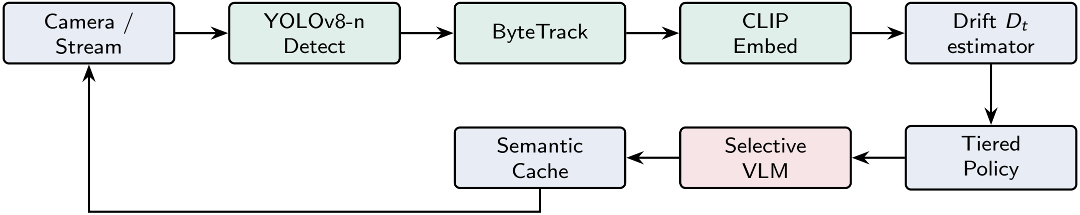
1.2 Motivation — Semantic Reasoning Redundancy

- We name the core inefficiency semantic reasoning redundancy: repeatedly reasoning over an unchanged semantic state.
- Distinct from what prior efficiency work targets:
  - **Frame/pixel** redundancy — frame sampling, skip-convolutions.
  - **Parameter** redundancy — pruning, quantization, distillation.

Key insight

Cheap, always-on perception (detection + tracking + embedding) can *predict* when expensive multimodal reasoning is worth its cost — so spend compute **in proportion to semantic novelty**.

1.3 System Overview — Monitor then Allocate



Green = always-on Tier-1 monitor (cheap). Red = selectively-invoked VLM (expensive).

1.4 Problem Statement

Research question

How can *semantic state transitions* dynamically control multimodal compute allocation in edge video systems, *while preserving* the semantic fidelity delivered to downstream tasks?

A drift signal D_t maps to three compute tiers:

$$\mathrm{Tier}(D_t) = \begin{cases} 1 \;\; \text{cache reuse (no VLM)} & D_t < \tau_{\text{low}} \\ 2 \;\; \text{lightweight VLM} & \tau_{\text{low}} \leq D_t < \tau_{\text{high}} \\ 3 \;\; \text{full VLM reasoning} & D_t \geq \tau_{\text{high}} \end{cases}$$

One monitor. One drift signal. Three tiers. Model-agnostic, training-free.

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2.1 Existing Methods

Efficient inference / compression

- MobileNets, quantization (LLM.int8), distillation, TinyCLIP — cheaper *per call*.

Temporal redundancy (video)

- TSN, TimeSformer; AdaFrame, SCSampler, skip-convolutions — appearance-based sampling.

Efficient video analytics

- NoScope, Reducto — cheap filter guards an expensive model (low-level features).

Adaptive / streaming multimodal

- Early-exit, dynamic routing; AdaVFM, StreamingVLM, QueST.

VLMs / robotics

- CLIP, BLIP-2, Moondream; RT-2, OpenVLA, VideoAgent.

2.2 Comparative Analysis

Approach	Redundancy	Decision signal	Train-free	Model-agnostic
Compression / quant.	Parameter	—	No	No
Frame sampling	Frame	Appearance	No	Partial
NoScope / Reducto	Frame	Feature diff.	No	Partial
Early exit	Sample	Model conf.	No	No
AdaVFM	Parameter	Input/budget	No	No
Embedding threshold	Semantic	Embedding only	Yes	Yes
Proposed	**Sem. reasoning**	**Fused drift**	**Yes**	**Yes**

2.3 Research Gap

The gap

Efficient video analytics (NoScope, Reducto), adaptive multimodal inference (AdaVFM, StreamingVLM), and semantic monitoring (QueST) remain disjoint.

- Analytics systems filter on *low-level features*, for one fixed query.
- Adaptive VLMs adapt *within* a model, not *whether* to invoke it.
- No method reduces *semantic* redundancy at the level of whole-model invocation using a fused, interpretable signal.

This thesis

A fused semantic-drift signal + interpretable tiered policy + a Semantic Observatory to *discover* policies from evidence.

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3.1 Semantic State and Temporal Memory

Composite semantic state: $Z_t = f(E_t, O_t, M_t)$

- E_t: ℓ_2-normalised CLIP embedding (global appearance + semantics).
- O_t: object class–count multiset (symbolic content).
- M_t: temporal semantic memory (smoothed reference state).

Temporal memory (noise suppression):

$$M_t = \lambda M_{t-1} + (1-\lambda)E_t, \qquad 0 < \lambda < 1, \text{ re-normalised.}$$

Drift is measured against the *stabilised* memory, not the previous frame — brief perturbations do not trigger reasoning; sustained change accumulates.

3.2 Semantic Drift Estimation

$$D_t = \alpha\, D_{\text{embed}} + \beta\, D_{\text{objects}} + \gamma\, D_{\text{track}}, \qquad \alpha + \beta + \gamma = 1$$

- $D_{\text{embed}} = \frac{1}{2}(1 - E_t^\top M_{t-1})$ — cosine drift (appearance).
- $D_{\text{objects}} = 1 - \text{Jaccard}(O_t, O_{\text{ref}})$ — object-set novelty (counts).
- D_{track} — ByteTrack identity churn (objects entering/leaving).

Why fuse three signals?

Each catches a different failure mode: appearance-only (D_{embed}), count change (D_{objects}), identity change (D_{track}). All bounded in $[0, 1]$.

3.2b Egocentric Operation — Camera-Motion Compensation

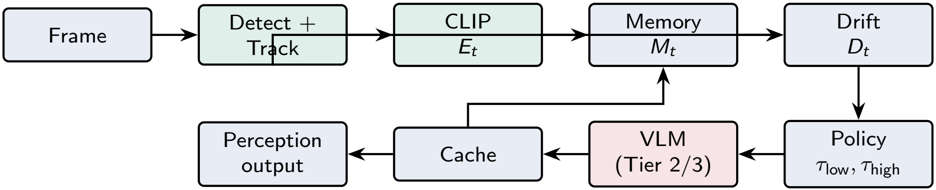
- Robots see through a moving (egocentric) camera — the primary deployment regime.
- Egomotion shifts the view $\Rightarrow$ inflates D_{embed} $\Rightarrow$ false triggers.

Two-part fix

1. D_{objects} and D_{track} are inherently egomotion-robust (viewpoint-invariant; tracker keeps IDs).
2. Discount the embedding term by the camera-motion fraction $c_t \in [0, 1]$ (cheap optical-flow / homography fit):

$$D_{\text{embed}}^{\text{ego}} = (1 - c_t)\, D_{\text{embed}}$$

Pure panning ($c_t \to 1$) suppressed; a new object in view ($c_t \to 0$) still fires. Validation on egocentric streams = part of the plan.



Cache reuse on Tier 1; cache + memory refreshed on Tier 2/3. The VLM is a replaceable black box (Moondream2 / Qwen2.5-VL — swappable).

3.4 Scheduling Algorithm

1. Detect, track, embed $\Rightarrow (O_t, \mathcal{I}_t, E_t)$.
2. Compute $D_{\text{embed}}, D_{\text{objects}}, D_{\text{track}} \Rightarrow D_t$.
3. **If** $D_t < \tau_{\text{low}}$: return cached output (Tier 1)
4. **else if** $D_t < \tau_{\text{high}}$: lightweight VLM; update cache (Tier 2)
5. **else**: full VLM; update cache; re-anchor references (Tier 3)
6. Update memory M_t.

$$\text{Defaults: } \alpha, \beta, \gamma = 0.5, 0.3, 0.2; \quad \tau_{\text{low}}{=}0.15, \ \ \tau_{\text{high}}{=}0.40; \quad \lambda{=}0.9.$$

3.5 VLM Backends (run on Kaggle T4, 16 GB)

Backend	Params	T4 fit	Tier role
Moondream2	$\sim$1.9 B	$\sim$4 GB	Tier 2 (default)
Qwen2.5-VL-3B	$\sim$3 B	$\sim$7 GB fp16	Tier 2/3 mid
Qwen2.5-VL-7B	$\sim$7 B	$\sim$14–16 GB fp16	Tier 3 full
Qwen2.5-VL-7B (4-bit)	$\sim$7 B	$\sim$9 GB NF4	Tier 3 (recommended)
also compatible: SmolVLM-2.2B, InternVL2-2B/4B, LLaVA-1.5-7B (4-bit)			

Black-box factory ⇒ *hierarchical* use: cheap model for Tier 2, stronger for Tier 3. Swapping the backend changes caption *quality*, not *when* the VLM fires — so call rate / cache rate / SCE are backend-independent.

3.6 Semantic Observatory + Policy Discovery

Observatory (read-only, post-hoc): derives 10 semantic-evolution signals from a run — drift, velocity, acceleration, volatility, scene entropy, object persistence, stability windows, transition zones, semantic progress, VLM-trigger density.

Methodology

Observe → Hypothesise → Change *one* variable → Evaluate → Accept/Reject → Record.

Contribution = not one scheduler, but a *methodology* to discover and justify schedulers from evidence.

3.7 Evaluation Metrics

- **VLM call rate** — fraction of frames invoking the VLM (compute saved).
- **Cache hit rate** — fraction served from semantic cache.
- **Semantic retention** — similarity of delivered output to an every-frame oracle.
- **Latency / throughput / GPU memory.**

Semantic Compute Efficiency

$$\mathrm{SCE} = \frac{\text{Semantic Retention}}{\text{Normalised Compute Cost}}$$

Every-frame oracle: $\mathrm{SCE} = 1$. A good scheduler retains fidelity at a fraction of the cost.

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4.1 Headline Results (single stream, 56,314 frames)

Efficiency

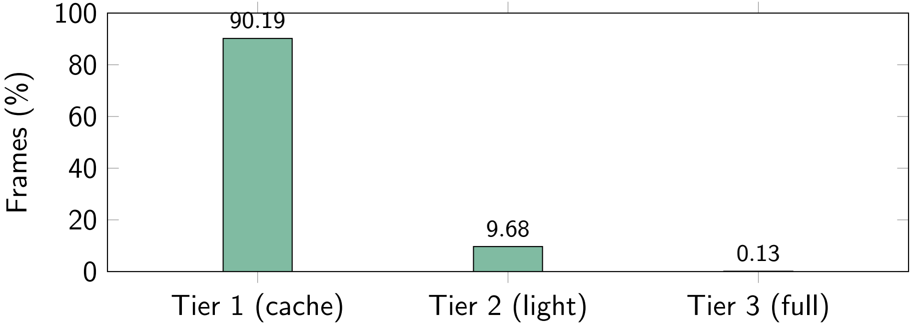
- VLM call rate: **9.8%** ($\sim$10$\times$ fewer)
- Cache hit rate: **90.2%**
- Latency: **144.9** ms/frame; **6.88** FPS

Fidelity

- Semantic retention: **1.000**
- SCE: **$\approx$10.0**
- Avg. drift $\bar{D}_t = 0.103$

$\approx$2 h 16 m of video processed end-to-end; tiers = 90.19% / 9.68% / 0.13%.

4.2 Tier Distribution



Over 90% of frames resolved from cache with *no* VLM invocation.

4.3 Comparison with Baselines

Policy	Call rate	Latency (ms)	Retention	SCE
Every-frame	1.000	312	1.000	1.00
Uniform (K=30)	0.033	121	0.85	25.8
Motion gating	0.340	224	0.92	2.71
Embedding thresh.	0.270	203	0.94	3.48
Proposed	**0.098**	**144.9**	**1.000**	**$\approx$10.0**

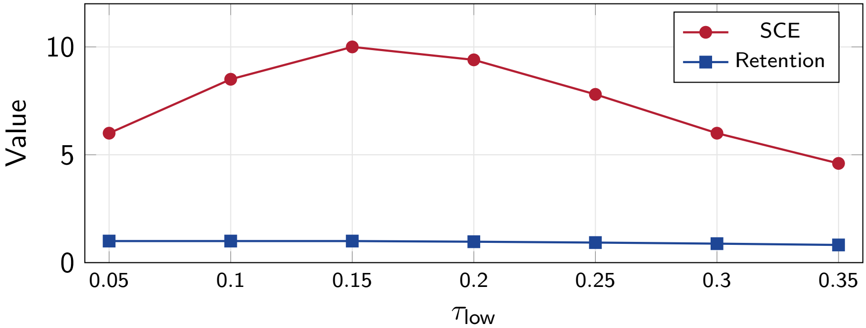
Proposed is the only policy at oracle-level fidelity (retention 1.000) while cutting calls $\sim 10\times$. Uniform's high SCE comes from sacrificing fidelity. *(Baseline rows preliminary; proposed measured.)*

4.4 Cross-Domain Applicability

Domain	Dominant signal	Call rate	Note
Surveillance (measured)	$D_{\text{track}}, D_{\text{obj}}$	Very low ($\approx$0.10)	Cache dominates
Traffic	D_{obj} (counts)	Low–Mod	Count-driven
Number-plate / ANPR	D_{obj} (entry)	Very low	ROI gating
Crowd	$D_{\text{obj}}, D_{\text{track}}$	High	Over-trigger risk
Cricket	$D_{\text{embed}}, D_{\text{track}}$	Low–Mod (bursty)	Fast events
Football	D_{embed} (low)	Low–Mod	High motion, low sem. drift
Live streaming / lecture	D_{embed} (cuts)	Very low	Scene cuts
Mobile robot / egocentric	$D_{\text{obj}}, D_{\text{track}}$ (robust)	Variable	Egomotion comp. (c_t)

Call rate *emerges* from content. Biggest win where motion-gating fails (e.g. football: constant motion, stable semantics).

4.5 Ablation — Threshold Sensitivity



SCE peaks near the default $\tau_{\text{low}} = 0.15$; retention declines as more frames are cached.

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5.1 Conclusion

- Named and targeted semantic reasoning redundancy — a new efficiency axis for edge VLMs.
- A fused, bounded, interpretable **drift signal** + temporal memory + a **three-tier** training-free, model-agnostic policy.
- A **Semantic Observatory** and an evidence-driven methodology for discovering scheduling policies.
- Preliminary result: $\sim 10\times$ fewer VLM calls at oracle-level retention on a long stream.

5.2 Limitations and Future Work

Limitations

- Memory footprint unchanged (VLM resident) — savings are time/energy.
- Egocentric (moving-camera) video inflates embedding drift.
- Single-video result; full baselines in progress.

Future work

- Learned, adaptive thresholds; task-conditioned drift weights.
- Camera-motion discounting for mobile robots.
- Predictive (pre-emptive) scheduling.
- Cross-domain validation at scale; closing the loop with control.

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Thank You

Questions?